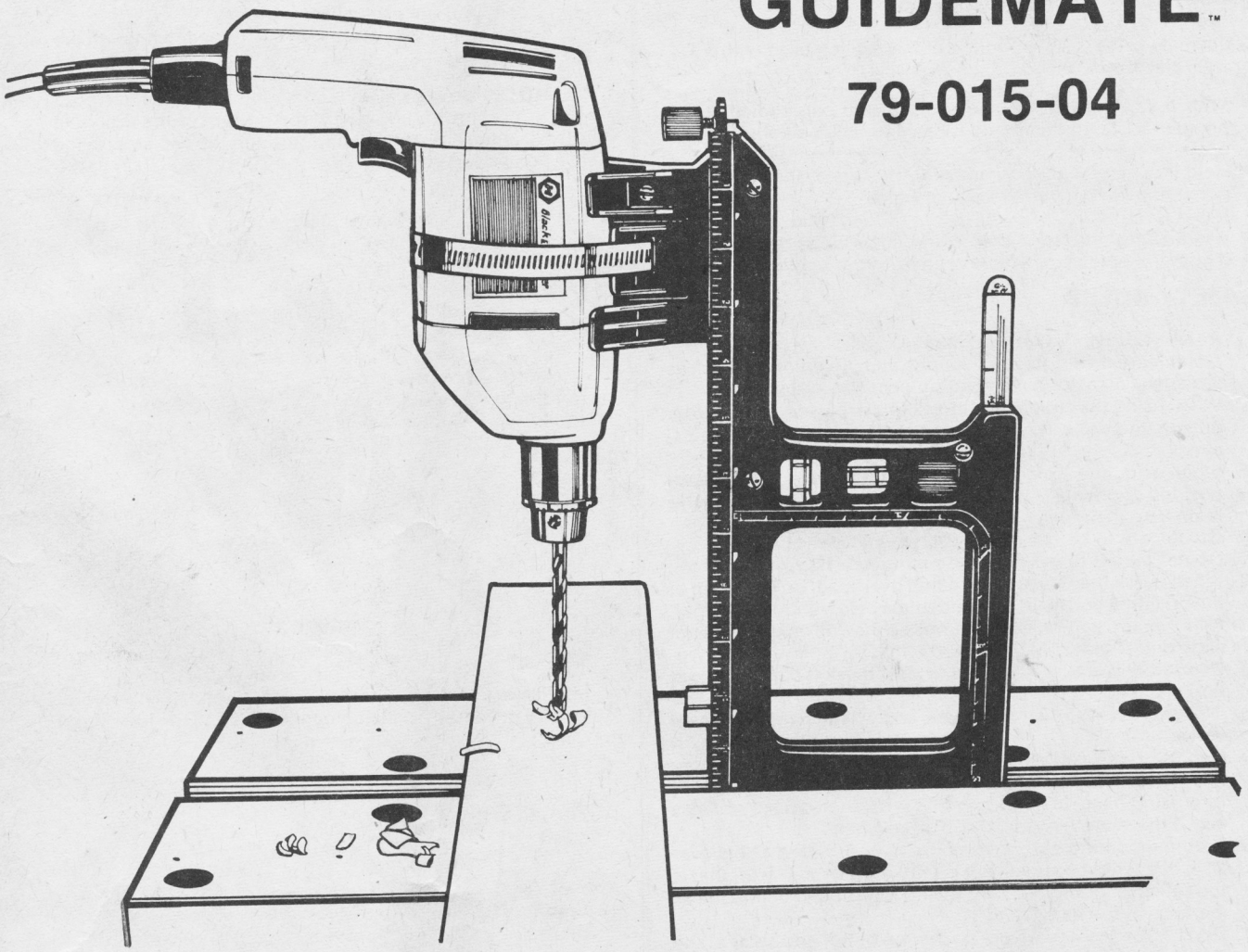
Black & Decker®

# Workmate® ACCESSORY

## GUIDEMATE™

## 79-015-04



Your new Workmate Accessory is a quality piece of equipment that has been carefully designed and manufactured to provide years of dependable service.

This Instruction Manual will familiarize you with your Accessory.

It lists important facts and features which describe your Workmate Accessory's construction and basic operating techniques.

Set-up instructions and various usage situations are also provided.

Read this manual carefully. Only by being totally aware of what your Workmate Accessory can do will you be able to put it to the numerous tasks it is capable of performing.

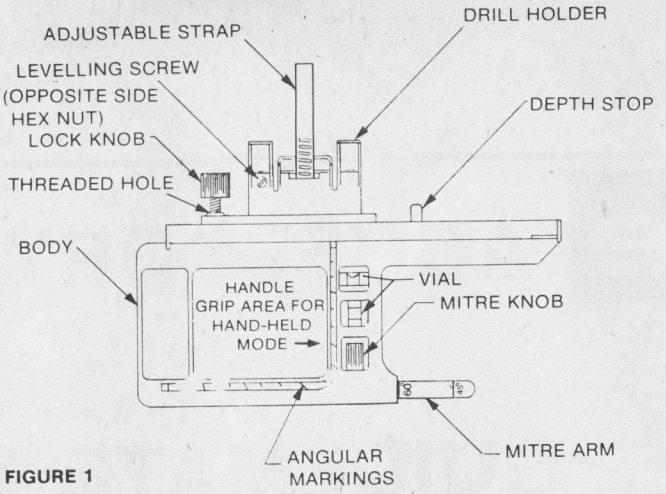



**FIGURE 1**

Guidemate performs a multitude of tasks when used with a 1/4" or 3/8" electric drill.

- With a Workmate it can be used to drill perpendicular and angular holes. In this mode it has a function similar to a drill press.
- Held free hand it adds accuracy to free hand drilling both for perpendicular as well as angular holes.
- It is a level.
- It is a 12" ruler in both metric and imperial units.
- It is a drill holder for grinding and buffing applications.

## SAFETY RULES

1. Read your Drill Owner's Manual carefully before using Guidemate. Learn its applications and limitations as well as the specific safety consideration peculiar to this tool.
2. When using the Guidemate check for wiring locations before drilling into walls, floor, etc. to avoid receiving an electrical shock.
3. When using the Guidemate in the Workmate as a drill press, position Guidemate so that you do not drill into the Workmate Vise Brackets.
4. Guidemate to be used with 1/4" or 3/8" Drills only.
5. Do not mount the Guidemate in a bench vise.
6. When using the Guidemate to drill overhead, do not let go of the drill until you have the Guidemate below shoulder height. At this point, tighten the lock knob so the drill holder does not slide out of the Guidemate body. Fig. 1.
7. Use safety glasses. Also face or dust mask if cutting operation is dusty.
8. Disconnect Drill when not in use, and when changing accessories.
9. Handle grinding wheels carefully to avoid bumping or dropping. Do not use a grinding wheel that has been dropped. Before using, inspect each grinding wheel for cracks or flaws and if these are evident, discard the wheel.
10. Before mounting a wire wheel brush or grinding wheel over 2" in diameter, be sure that it is marked with an R.P.M. that is the same as, or higher than, the no load speed of the drill as marked on the nameplate.
11. Don't force the drill. It will do the job better and safer at the rate for which it was designed.
12. When drilling light gauge metal, fasten the piece down so that it does not spin.

## SETUP INSTRUCTIONS

To use the Guidemate the following setup steps are required to be performed.

1) Drill is mounted in the drill holder.
2) Drill is levelled with respect to the drill holder base.
3) Depth stop is inserted into the Guidemate body.
4) Drill with its holder are inserted in the Guidemate body. Direction of the drill is governed whether the Guidemate is to be used with a Workmate or in the hand held mode of operation.

Figure 1 shows the basic parts of your Guidemate. Study Figure 1 before going through the setup procedure below.

## DRILL HOLDER SETUP

(a) Open the plastic bag with the Guidemate hardware and the bag should contain the following items:

| | |
|---|---|
| 1. Adjustable Strap | Qty. 1 |
| 2. #8-32 Levelling Screw | Qty. 1 |
| 3. #8 Hex Nut | Qty. 1 |
| 4. Lock Knob | Qty. 1 |
| 5. Depth Stop | Qty. 1 |

(b) Take the #8-32 screw, #8 hex nut and assemble on the drill holder in location shown in Fig. 1. Leave the screw loose at this point.

**NOTE:** Two flats are provided on the drill holder on one side to trap the hex nut. Check that the screw and nut are assembled correctly.

(c) Screw in the lock knob into the threaded hole on the drill holder. Fig. 1.

**NOTE:** Do not have the threaded section of the lock knob protruding below the bottom of the drill holder.

(d) Slide the adjustable strap through the two slots in the drill holder. Fig. 2.

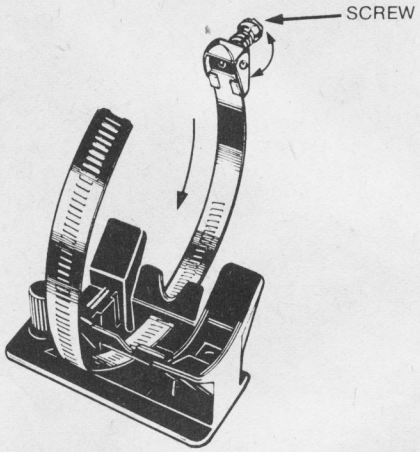



**FIGURE 2**

(e) Place your drill in the drill holder so that it is horizontal and the drill chuck is facing opposite end to the lock knob. Fig. 3.

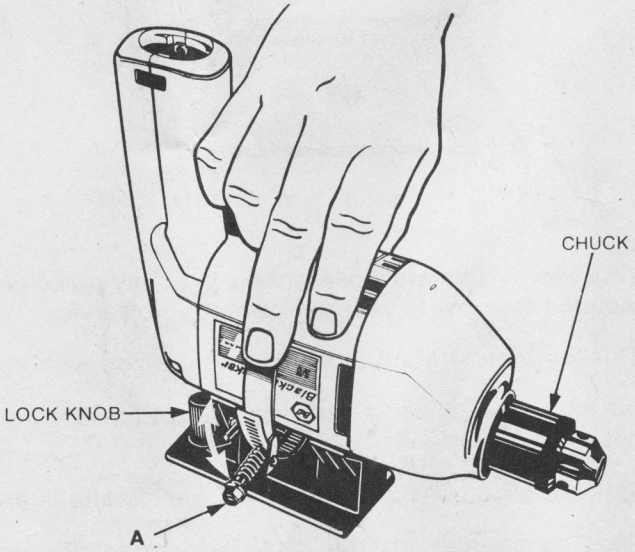



**FIGURE 3**

Clamp the drill lightly with the adjustable strap as shown in Fig. 4.

**CAUTION:** Do not cover drill air vents with the adjustable strap.

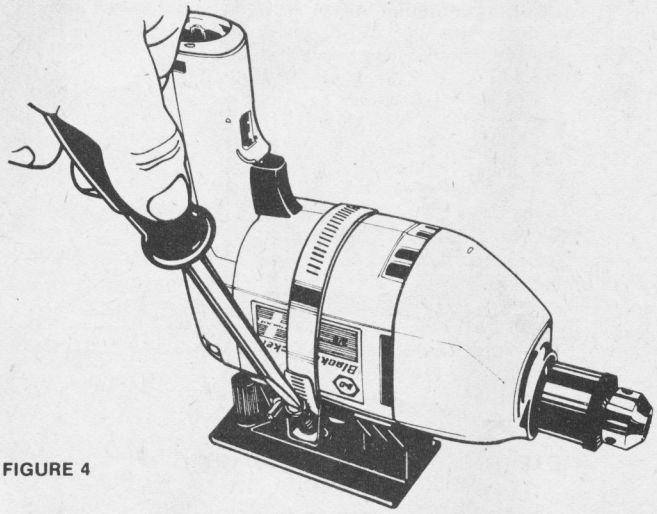


FIGURE 4

## 2. METHOD OF LEVELLING ELECTRIC DRILL IN HOLDER

(a) Install strap through slots in drill holder.

(b) Place drill holder and strap on level surface such as table top.

(c) Place drill bit in chuck of drill and position drill on holder to be approximately level.

(d) With screwdriver, tighten strap just enough to remove slack , so that the strap sits on drill casing.

(e) With straight edge, measure along the length of the drill bit to be sure that the drill is parallel to the level surface. Adjust levelling screw as required. Ref: "X" & "Y" dimensions must be equal.

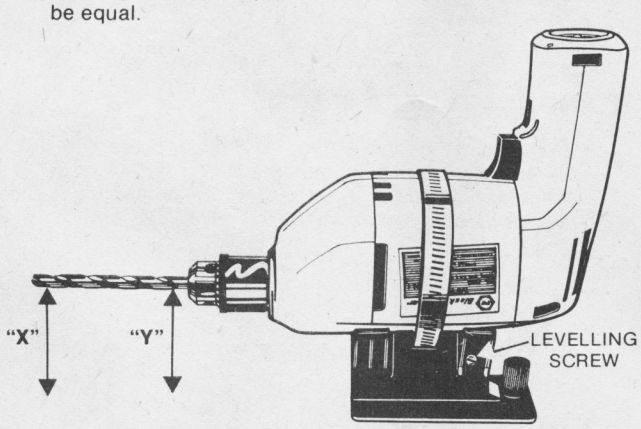



FIGURE 5

(f) By adjusting the levelling screw, and the strap tension screw, bring the drill to level and tighten both screws as required to maintain level position and hold drill securely.

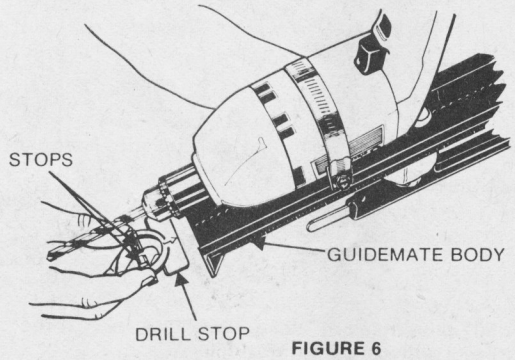



FIGURE 6

## 3. DEPTH STOP SETUP

(a) Insert the depth stop into the slot in the Guidemate body Fig. 6 by squeezing the depth stop arms at the same time sliding the depth stop in the slots of the body.

(b) **CAUTION:** The arrow on the depth stop **must always** point in the **opposite** direction that the drill bit is pointing.

(c) The depth stop can be positioned along the length of the Guidemate in 1/8" increments. To provide an incremental depth adjustment during drilling.

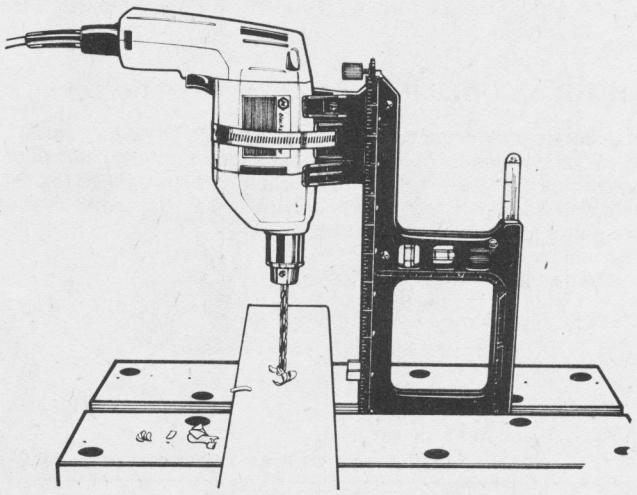


FIGURE 7

## 4. ASSEMBLY OF HOLDER TO BODY

If the Guidemate is to be used with a Workmate, insert the drill with its holder as shown Fig. 7.

Used in the hand held mode the drill and holder are inserted in the opposite direction. The depth stop is also reversed as shown in Fig. 12.

Once the drill and holder are inserted in the Guidemate body, tighten the lock knob. This secures the two parts together.

## 5. PERPENDICULAR DRILLING

(a) Unplug the drill.

(b) Open the Workmate vise jaws enough to clear the Guidemate body between the vise jaws as shown in Fig. 8.

**CAUTION:** Position the Guidemate away from the steel vise brackets.

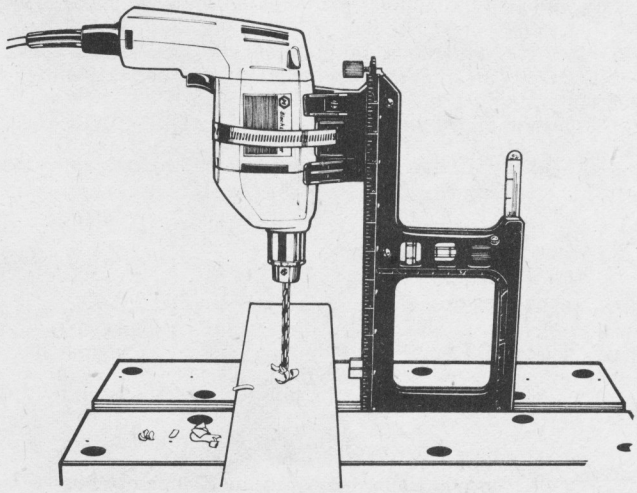


FIGURE 8

(d) Rest end of slide on vise jaw top, then line up the 90° marking on the Guidemate body Fig. 8 with the top face of the Workmate vise jaw.

(e) Now close the vise jaws parallel and tighten securely on the Guidemate body. Do not over tighten.
(f) Position the work to be drilled below the bit.
(g) Plug the drill in.
(h) Loosen the lock knob, start the drill. Guide the drill through the work piece.

**NOTE:** If you want to drill a hole to a depth, see depth stop setup. #3.

If you find in drilling you can not hold your workpiece securely, a Black & Decker Gripmate 79-011 horizonal clamp could do the job for you. (Gripmate not included with the Guidemate).

## ANGULAR DRILLING

Guidemate provides a method of drilling angular holes from 90° to 45° in increments of 5°. For proper angular drilling a 90° pilot hole about 1/8" deep (see Fig. 9 & 10) has to be predrilled to prevent drill wander. Pilot hole size can be the same size drill as you are going to use to drill the angular holes.

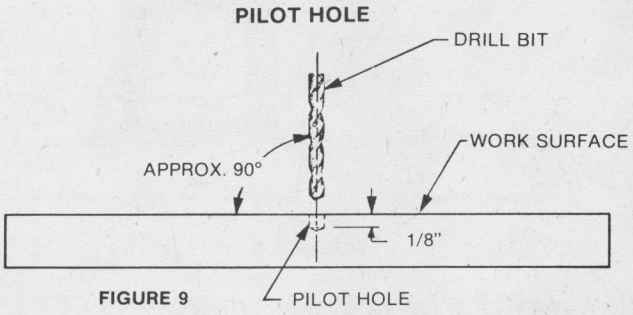



**FIGURE 9**

## ANGULAR DRILLING USING PILOT HOLE

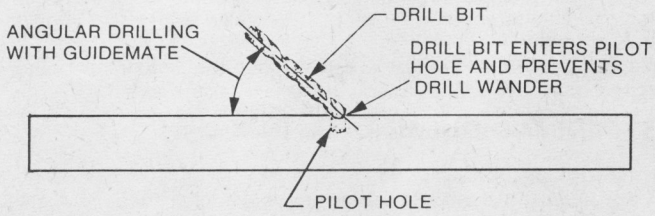



**FIGURE 10**

(a) After predrilling the pilot hole, setup the Guidemate in the vise jaws using the angular markings on the Guidemate body. Fig. 11.
(b) Place the workpiece on the vise jaw surface and clamp C-clamps or Gripmate (not included) after aligning with the drill bit.
(c) Drill as in above procedure.

**NOTE:** If you want to drill a hole to a depth, see depth stop setup #3.

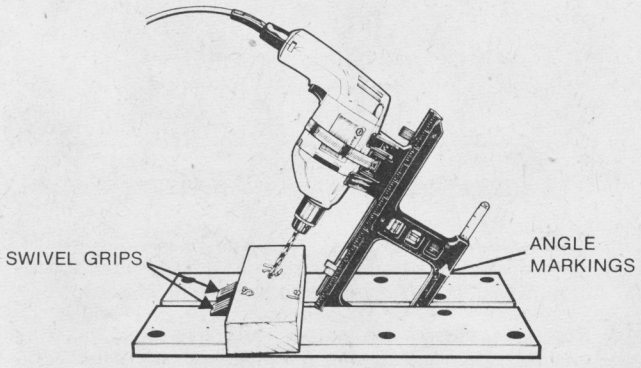



**FIGURE 11**

## 6. GUIDEMATE USED FREE HAND FOR DRILLING

For free hand drillling, the drill and holder are positioned as shown in Fig. 12.
To drill perpendicular holes follow the below procedure.

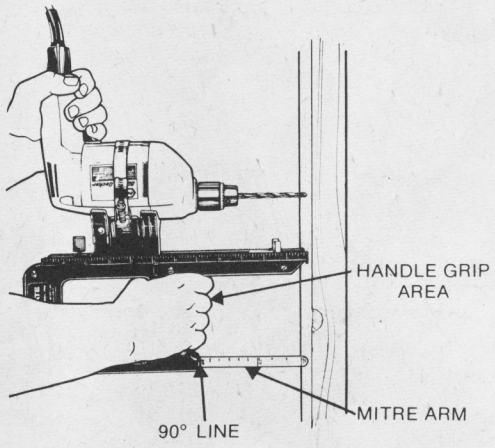



**FIGURE 12**

## I PERPENDICULAR HOLES

(a) Locate the mitre arm at 90° marking and tighten the mitre knob securely to prevent slippage.
(b) With the drill and holder secured to the Guidemate body using the lock knob position the drill bit for drilling as shown in Fig. 12. Note position of hands.
(c) Press firmly to prevent the Guidemate from moving. Loosen the lock knob and complete the drilling. Do not force the drill.
(d) Tighten the lock knob after drilling.

## II ANGULAR HOLES

In the hand held mode (Fig. 13) angular holes from 90° to 45° are possible in 5° increments.
Angular holes differ in setup from the perpendicular hole setup procedure in two ways.

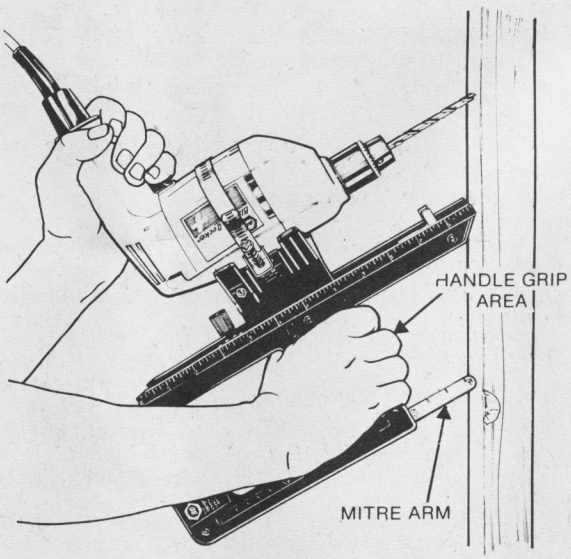



**FIGURE 13**

(a) A 90° pilot hole 1/8" deep, see Fig. 9 & 10, is required to prevent drill wander during drilling.

(b) Angles are achieved by adjusting the mitre arm to the required angle as stamped on the mitre arm. The mitre arm is tightened using the mitre knob.
(c) Drilling proceeds in a similar manner to perpendicular drilling.

**CAUTION:** When using the Guidemate free hand, check wiring location before drilling to avoid an electrical shock.

## 7. HORIZONTAL DRILL STAND

**CAUTION**

Always position the holder and the drill with the chuck pointing to the right as shown Fig. 14. This causes the face of the wheel closest to the operator to rotate downward and throw most sparks and particles also downward.

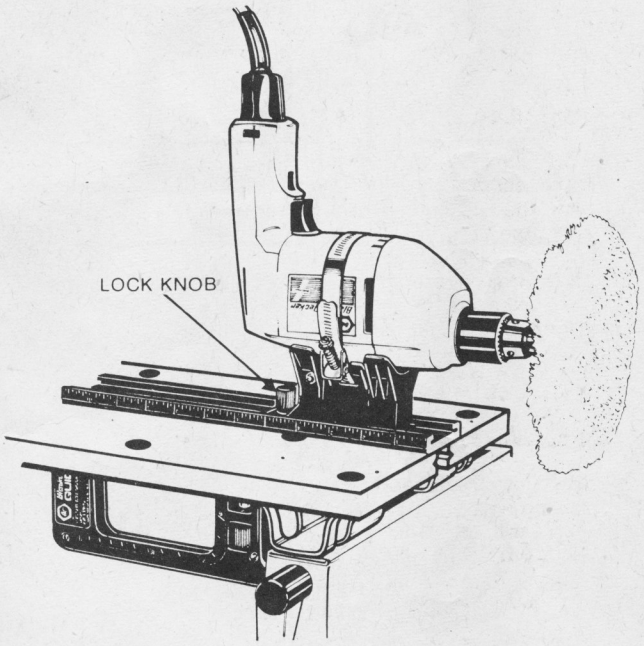



FIGURE 14

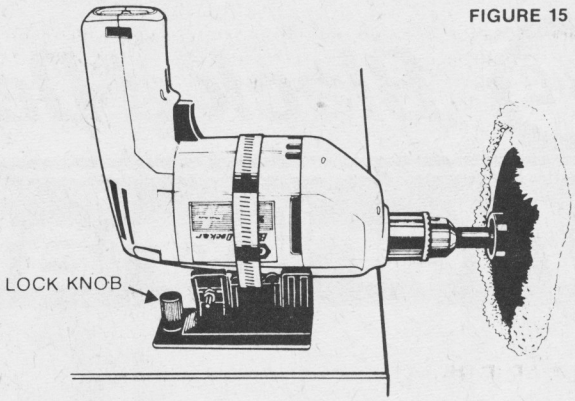
### I Holder Fastened to Work Bench

(a) Back off lock knob so the thread is not protruding below the bottom of the drill holder.
(b) Remove the drill from the drill holder and the adjustable strap.
(c) Position the holder in the location you would like to work in. The lock knob position always to your left.
(d) Two #10 screw clearance holes provided in the drill holder. Fig. 16.
(e) Screw the holder down to your work bench with two #10 screws or bolts (not included).
(f) Remount your drill on the holder and tighten the strap.

**NOTE:** Drill does not have to be levelled for this operation, just strapped down firmly with the drill chuck positioned at the opposite end to the lock knob. Fig. 15.

Before mounting any accessories in your drill, read the following sections on Recommended Accessories, Grinding, Wire Brushing, and Buffing Instructions.

FIGURE 15



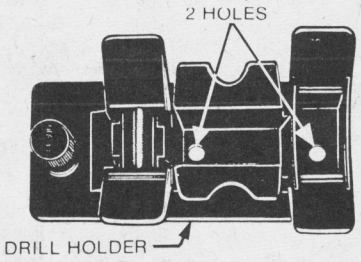




FIGURE 16

### II Recommended Accessories

The following accessories are recommended for use when your drill is mounted in the horizontal stand.

**Caution:** The use of any other accessory might be hazardous. Carefully read instructions provided with accessory and safety rules.

Wheel Arbor — Up to 2" diameter.
Wheel Arbor — 1/4" shank, 1/2" body.
Grinding Wheels — Up to 2" diameter.
Wire Wheel Brushes — Up to 4" diameter.
Buffing Wheels — Up to 4-5/8" diameter.
Sanding Discs — Up to 5" diameter.
Polishing Bonnets — Up to 5" diameter.
Flexible Shaft — 1/4" shank and 1/4" chuck.

### III Grinding Instructions

Put on safety glasses or other eye protection and hold the Work firmly. Feed the work slowly into the wheel at the desired grinding angle. Treat the wheel with respect . . . do not jam the work into the wheel or use unnecessary pressure. Grind only on the face of the wheel, unless you have a special wheel specifically made to permit grinding on the side of the wheel. When grinding tempered tools such as chisels and knives, keep a container of cutting fluid or water nearby. During the grinding operation, frequently dip the hot part of the work into the liquid to avoid overheating and losing the temper. In fact, all metals can be ground more efficiently if they are "quenched" in this manner to avoid overheating which can soften or "burn" the metal.

### IV Wire Brushing Instructions

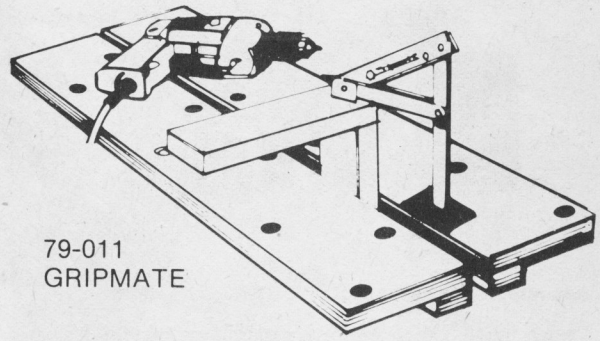
Wire Brushing quickly removes rust, scale or old paint. Always wear safety glasses or other eye protection when using wire wheels as pieces of wire may break off and be thrown from the rotating brush. Handle Wire Wheel Brushes carefully with gloves or rags to avoid puncturing the skin.

### V Buffing Instructions

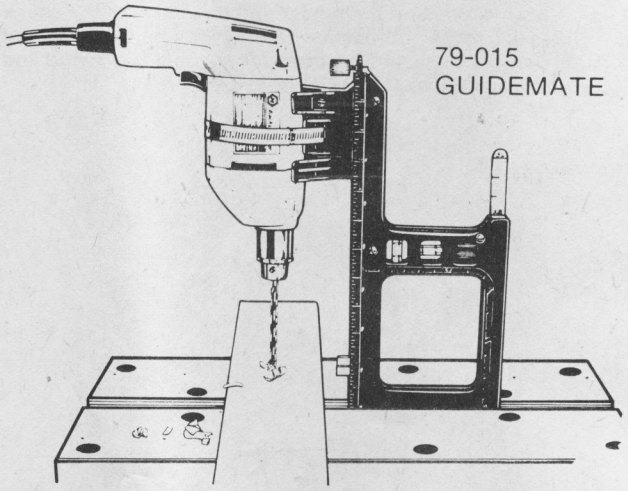
Use cotton Buffing Wheels, normally with stick buffing compound applied to the face of the wheel (usually about 5/8" wide). Wear safety goggles for protection against particles of compound thrown off by the wheel. Do not jam work into wheel. Hold work firmly.

# Black & Decker®

# Workmate® ACCESSORIES

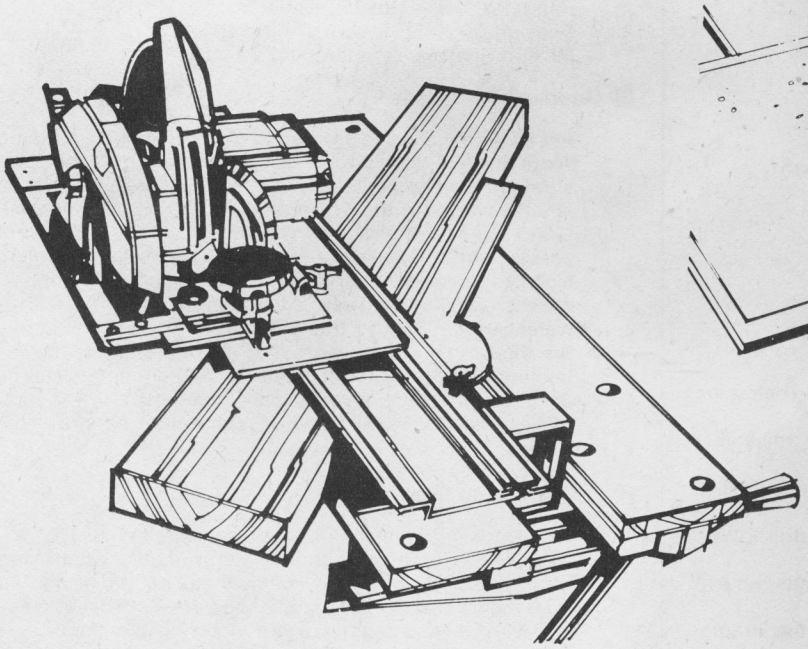


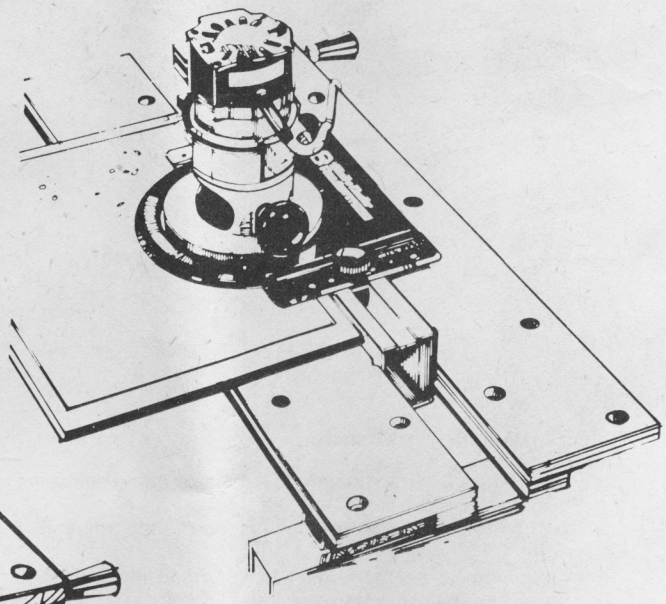
79-011
GRIPMATE



79-015
GUIDEMATE

79-012 MITREMATE





79-013 ROUTERMATE

**Black & Decker Mfg. Co. Ltd., Brockville, Ontario, Canada**

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